

CUET PG (SCQP09) PRACTICE QUESTION BANK

Computer Science & Information Technology | Subject: Data Structures

**TOPIC: QUEUES (EASY
LEVEL)**

50 MCQs

Instructions:

- Each question carries 4 marks. For each incorrect response, 1 mark will be deducted (-1).
- Select the single best answer choice for each multiple-choice question.
- Answer Key is provided on the final page of this question sheet.

Q1. A Queue works on which of the following principle?

- (A) LIFO (Last In First Out) (B) FIFO (First In First Out)
(C) LILO (Last In Last Out) (D) Both (B) and (C)

Q2. The operation of adding an element into a queue is called:

- (A) Push (B) Enqueue
(C) Pop (D) Dequeue

Q3. The operation of removing an element from a queue is known as:

- (A) Enqueue (B) Dequeue
(C) Peek (D) Pop

Q4. From which end are elements inserted into a standard linear queue?

- (A) Front end (B) Rear end
(C) Middle position (D) Top end

Q5. From which end are elements deleted from a standard linear queue?

- (A) Rear end (B) Front end
(C) Base end (D) Any position

Q6. Trying to delete an element from an empty queue causes which condition?

- (A) Queue Overflow (B) Queue Underflow
(C) Memory Leak (D) Garbage Collection

Q7. Trying to insert an element into a completely full queue causes:

- (A) Queue Underflow (B) Queue Overflow
(C) Null Pointer Exception (D) Infinite Loop

Q8. Initial values of FRONT and REAR pointers in an array-based implementation for an empty queue are usually set to:

- (A) FRONT = 0, REAR = 0 (B) FRONT = -1, REAR = -1
(C) FRONT = 0, REAR = -1 (D) FRONT = 1, REAR = 1

Q9. What is the time complexity of Enqueue operation in a standard queue implemented using an array with a rear pointer?

- (A) $O(1)$ (B) $O(N)$
(C) $O(\log N)$ (D) $O(N^2)$

Q10. What is the main drawback of a standard linear array implementation of a queue?

- (A) Elements cannot be accessed (B) Memory space gets wasted as unused slots before FRONT cannot be reused
(C) Takes $O(N^2)$ time for insertion (D) Dynamic size allocation is impossible

Q11. In a circular queue of size N, if REAR is at position i, what is the next position of REAR after an Enqueue?

- (A) $(\text{REAR} + 1) \% N$
- (B) $(\text{REAR} + 1)$
- (C) $\text{REAR} \% N$
- (D) $(\text{REAR} - 1) \% N$

Q12. Statement I: In a Queue, the insertion takes place at the Rear end.

Statement II: In a Stack, insertion and deletion take place at different ends.

- (A) Both Statement I and II are true
- (B) Both Statement I and II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Q13. In a circular queue of capacity N (using 0 to N-1 indexing), what is the condition for Queue Full if one position is left empty?

- (A) $(\text{REAR} + 1) \% N == \text{FRONT}$
- (B) $\text{REAR} == \text{FRONT}$
- (C) $\text{REAR} == N - 1$
- (D) $\text{FRONT} == (\text{REAR} + 1)$

Q14. In a circular queue implemented using an array, when is the queue considered EMPTY?

- (A) $\text{REAR} == N - 1$
- (B) $\text{FRONT} == -1$ and $\text{REAR} == -1$
- (C) $\text{FRONT} == \text{REAR} + 1$
- (D) $\text{FRONT} == N$

Q15. Which real-world scheduling model directly utilizes a Queue data structure?

- (A) Undo/Redo operations in text editor
- (B) First-Come, First-Served (FCFS) CPU scheduling
- (C) Recursion call evaluation
- (D) Expression syntax checking

Q16. Match List I with List II:

List I: (P) Enqueue, (Q) Dequeue, (R) Peek, (S) Circular Queue

List II: (1) View front element, (2) Solves linear queue space wastage, (3) Insert element, (4) Delete element

- (A) P-3, Q-4, R-1, S-2
- (B) P-4, Q-3, R-1, S-2
- (C) P-3, Q-4, R-2, S-1
- (D) P-1, Q-2, R-3, S-4

Q17. What data structure is used to implement Breadth First Search (BFS) graph traversal?

- (A) Stack
- (B) Queue
- (C) Tree
- (D) Binary Search Tree

Q18. What is a Deque (Double Ended Queue)?

- (A) Queue where insertion and deletion happen only at center
- (B) Queue where insertion and deletion can occur at both ends
- (C) Queue implemented using two stacks
- (D) Queue with priority attached to elements

Q19. In a Priority Queue, how are elements processed/dequeued?

- (A) Strictly based on insertion arrival time
- (B) Based on priority value associated with elements
- (C) Randomly selected element is removed
- (D) Elements with lowest memory address are popped first

Q20. When a queue is implemented using a singly linked list with head and tail pointers, where does Enqueue occur?

- (A) At the head node (front)
- (B) At the tail node (rear)
- (C) In the middle node
- (D) At a random position

Q21. Referring to Q20, where does Dequeue occur in a singly linked list queue to ensure O(1) time complexity?

- (A) At the head node (front)
- (B) At the tail node (rear)
- (C) At any random node
- (D) At the node with largest value

Q22. What is the time complexity of Dequeue operation in a linked list implementation of a queue with a head pointer?

- (A) $O(1)$
- (B) $O(N)$
- (C) $O(\log N)$
- (D) $O(N^2)$

Q23. Given an empty queue, the following operations are performed: Enqueue(10), Enqueue(20), Dequeue(), Enqueue(30), Dequeue(). What is the element at Front now?

- (A) 10
- (B) 20
- (C) 30
- (D) Queue is empty

Q24. An initial empty queue undergoes operations: Enqueue(A), Enqueue(B), Enqueue(C), Dequeue(), Enqueue(D). Which element is deleted next on Dequeue()?

- (A) A
- (B) B
- (C) C
- (D) D

Q25. Statement I: Priority Queue can be implemented using a Max-Heap or Min-Heap.

Statement II: In a Max-Heap implementation of Priority Queue, the highest priority element is always at the root.

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true, Statement II is false
- (D) Statement I is false, Statement II is true

Q26. What is an Input-Restricted Deque?

- (A) Insertion allowed at only one end, deletion allowed at both ends
- (B) Deletion allowed at only one end, insertion allowed at both ends
- (C) Insertion and deletion allowed at front only
- (D) Insertion and deletion restricted at both ends

Q27. What is an Output-Restricted Deque?

- (A) Insertion allowed at one end, deletion at both ends
- (B) Deletion allowed at only one end, insertion allowed at both ends
- (C) Deletion restricted to center element
- (D) No deletion allowed

Q28. In a circular queue of size 5, if FRONT = 2 and REAR = 4, how many elements are currently present in the queue?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q29. In a circular queue of size 6, if FRONT = 4 and REAR = 1, how many elements are present in the queue?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q30. Formula to count total number of elements in a Circular Queue of capacity N (given FRONT and REAR indices, when queue is not empty):

- (A) $(\text{REAR} - \text{FRONT} + N) \% N + 1$
- (B) $\text{REAR} - \text{FRONT}$
- (C) $(\text{REAR} + \text{FRONT}) \% N$
- (D) $N - \text{FRONT} + \text{REAR}$

Q31. Which data structure is used by printer spoolers to hold print jobs in order of arrival?

- (A) Stack
- (B) Queue
- (C) Binary Tree
- (D) Graph

Q32. In a Queue implemented using an array, if elements are deleted from Front without shifting elements left, what happens to Front pointer?

- (A) Increments towards Rear
- (B) Decrements towards 0
- (C) Remains unchanged
- (D) Resets to -1

Q33. Minimum number of Stacks needed to implement a Queue is:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q34. What operation is performed when inserting an element into a Priority Queue implemented using a Min-Heap?

- (A) Heapify / Up-Heapify
- (B) Left rotation
- (C) Linear search
- (D) Pop from top

Q35. What is the time complexity of Enqueue and Dequeue operations in a Binary Heap based Priority Queue?

- (A) $O(1)$ for both
- (B) $O(\log N)$ for both
- (C) $O(N)$ for both
- (D) $O(N^2)$ for both

Q36. What happens if you perform a Dequeue operation on a Queue when $FRONT == -1$?

- (A) Overflow Error
- (B) Underflow Error
- (C) Successful deletion
- (D) Queue size doubles

Q37. Which of the following data structures is NOT a linear data structure?

- (A) Queue
- (B) Stack
- (C) Binary Tree
- (D) Array

Q38. In a circular queue initialized with size 5, elements 10, 20, 30 are enqueued. What are current values of FRONT and REAR (0-indexed)?

- (A) $FRONT = 0$, $REAR = 2$
- (B) $FRONT = 0$, $REAR = 3$
- (C) $FRONT = 1$, $REAR = 3$
- (D) $FRONT = -1$, $REAR = 2$

Q39. Statement I: A queue can be used to synchronize asynchronous data transfer between two processes.

Statement II: Pipes in Unix operating systems use FIFO queue mechanism.

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true

Q40. If elements A, B, C, D are inserted into a queue in that order, what is the order of elements removed?

- (A) D, C, B, A
- (B) A, B, C, D
- (C) B, C, D, A
- (D) C, D, A, B

Q41. A double-ended queue (Deque) allows:

- (A) Insertion at front, deletion at rear only
- (B) Deletion at front, insertion at rear only
- (C) Insertion and deletion at both front and rear
- (D) Access only to middle element

Q42. Peek() operation in a queue returns:

- (A) The element at FRONT without removing it
- (B) The element at REAR without removing it
- (C) The element at FRONT and removes it
- (D) Total number of elements in queue

Q43. In a linked list representation of a queue, when the last remaining element is dequeued, what should FRONT and REAR be set to?

- (A) $FRONT = NULL$, $REAR = NULL$
- (B) $FRONT = Head$, $REAR = Head$
- (C) $FRONT = 0$, $REAR = 1$
- (D) Unchanged

Q44. Which algorithm design paradigm uses a Queue in Breadth First Search?

- (A) Dynamic Programming
- (B) Graph Traversal / Level Order
- (C) Divide and Conquer
- (D) Backtracking

Q45. If the size of an array implementing a linear queue is N, maximum number of elements that can be stored is:

- (A) $N - 1$
- (B) N
- (C) $N + 1$
- (D) $2N$

Q46. What is the space complexity of a Queue storing N elements?

- (A) $O(1)$
- (B) $O(N)$
- (C) $O(N^2)$
- (D) $O(\log N)$

Q47. In a queue implemented using a circular array of size 4, if FRONT = 0 and REAR = 3, performing one more Enqueue will result in:

- (A) Successful insertion at index 0 (B) Queue Overflow
(C) Queue Underflow (D) System Crash

Q48. Which among the following is TRUE about a Circular Queue?

- (A) It allows elements to be served in LIFO order (B) It avoids memory wastage of linear queues by connecting last position back to first
(C) It requires double memory space compared to linear queue (D) Insertion and deletion take $O(N)$ time

Q49. In CPU Round Robin Scheduling, ready processes are maintained in which data structure?

- (A) Stack (B) Circular Queue
(C) Binary Search Tree (D) Max-Heap

Q50. Level Order Traversal of a Binary Tree is implemented using:

- (A) Queue (B) Stack
(C) Priority Queue (D) Graph

ANSWER KEY (Q1 - Q50)

Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans	Q.	Ans
1	D	11	A	21	A	31	B	41	C
2	B	12	C	22	A	32	A	42	A
3	B	13	A	23	C	33	B	43	A
4	B	14	B	24	B	34	A	44	B
5	B	15	B	25	A	35	B	45	B
6	B	16	A	26	A	36	B	46	B
7	B	17	B	27	B	37	C	47	B
8	B	18	B	28	B	38	A	48	B
9	A	19	B	29	B	39	A	49	B
10	B	20	B	30	A	40	B	50	A